MIL-P-38027A(USAF)
12 November 1973
SUPERSEDING
MIL-P-38027(USAF)
5 October 1962

MILITARY SPECIFICATION

PRESSURE INDICATING SYSTEM, HYDRAULIC A/A27J-1

1. SCOPE

1.1 This specification covers one type of hydraulic pressure indicating system, designated A/A27J-1

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Boxes, Shipping, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-T-5350	Transmitters, Synchro Operated, Aircraft, General Specification For
MIL-H-5606	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, And Ordnance
MIL-I-6181	Interference Control Requirements Aircraft Equipment
MIL-I-7057	Indicators, Synchro, Aircraft, General Specification For
MIL-L-27160	Lighting, Instrument, Integral, White General Specification For

STANDARDS

Federal

FED-STD-595	Colors

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Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-143	Standards And Specifications, Order Of Precedence For The Selection Of
MIL-STD-785	Reliability Program For Systems And Equipment Development And Production
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods
MS28005	Transmitters, Pressure, Synchro, Aircraft
MS33639	Case, Instrument, Clamp-Mounted, Aircraft
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of the specifications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

FEDERAL AVIATION AUTHORITY

Technical Standard Order

TSO-C47	Pressure Instruments - Fuel, Oil And Hydraulic

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington DC 28402.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing (see 4.2.1).

3.2 Components. The system shall consist of:

a. One each Transmitter, Pressure, Hydraulic TRK-56/A27J-1.

b. One each Indicator, Pressure, Hydraulic AGK-10/A27J-1.

3.3 General specifications. The requirements of MIL-T-5350 and MIL-I-7057 apply as requirements of this specification with the exceptions and additions specified herein. When the requirements of the general specifications and this specification conflict, this specification shall govern.

3.3.1 Federal Aviation Agency TSO conformance. The TSO-C47 requirements apply as part of this specification.

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3.3.1.1 Federal Aviation Agency TSO application. The application procedure is presented in "Regulations of the Administrator, Subchapter C, Subpart A, paragraph 514.2 - TSO Authorization."

3.4 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.5 Design and construction.

3.5.1 System.

* 3.5.1.1 Reliability. The contractor shall establish a reliability program in accordance with MIL-STD-785. The acceptable mean-time-between failures (MTBF) shall be 1500 hours at a 90 percent confidence level.

* 3.5.1.2 Maintainability. The system shall be so designed and constructed that adjustments and repairs can be easily made by the personnel of operating units and overhaul bases, and be accomplished with tools ordinarily possessed by an instrument repairman.

3.5.1.3 Radio noise suppression. The system radio noise suppression requirements shall be in accordance with MIL-I-6181 for the frequency range of 0.15 to 150 megacycle (mc).

3.5.2 TRK-56/A27J-1 transmitter.

3.5.2.1 Hermetically sealed case. The case shall provide a hermetically sealed inclosure for the entire internal mechanism of the transmitter. The case shall be so designed that the internal mechanism may be removed, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures, unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that will be affected by the action of any atmosphere to which the transmitter may be subjected.

3.5.2.2 Case pressure release. A metal blowout device to discharge excessive case pressure shall be provided in the pressure transmitter. The device shall release the internal pressure through a 0.049-inch maximum diameter orifice when the internal pressure reaches 80 ±25 pounds per square inch (psi). The device shall not affect hermetic sealing.

3.5.2.3 Outline dimensions. The outline dimensions of the transmitter shall conform to MS28005-5, except that the electrical receptacle provided shall be designed for hermetic sealing. The pin arrangement and dimensions of the receptacle shall be in accordance with MS33678C-14S-2P.

3.5.2.4 Hydraulic connection. The transmitter shall be designed for straight-thread hydraulic connection as shown on MS28005-5. The pressure boss and the optional vent boss, if used, shall be identified with permanently marked, white letters P and V, 1/8 inch in height.

3.5.2.5 Overpressure stop. An overpressure stop shall be provided and shall be designed to restrain the prime mover. The stop shall allow the pointer of an indicator to travel beyond full scale to a point approximately midway between the last and the zero graduation.

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3.5.2.6 Weight. The weight of the transmitter shall not exceed 1.5 pounds.

3.5.3 AGK-10/A27J-1 indicator.

3.5.3.1 Case. The case shall be hermetically sealed, and the dimensions shall conform to MS33639 for the 1-inch diameter case. Dimension L shall be 2.75 ±0.25 inch.

3.5.3.1.1 A separate nonhermetically sealed cap may be used on the back of the case for installing the electrical receptacle. The cap shall be made of nonferrous, low-density metal.

3.5.3.1.2 The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/16 inch.

3.5.3.2 Integral lighting. The lighting of the dial and pointer shall be in accordance with MIL-L-27160 and as specified herein.

3.5.3.3 Pointer. The pointer shall be compatible with integral lighting and in accordance with Figure 1.

* 3.5.3.4 Dial. The dial shall conform to Figure 1. A minimum diameter of 15/16 inch shall be maintained across the outside ends of the graduations. The graduation for 2,000 psi shall be located on the horizontal centerline at the left side of the dial. The graduations for each 250 psi increment shall be 20 degrees apart with a tolerance of ±0 degrees 15 minutes for each graduation when referenced from the zero psi graduation.

3.5.3.4.1 Dial markings. Dimensions and colors of the dial and markings shall be as follows. The colors shall conform to FED-STD-595.

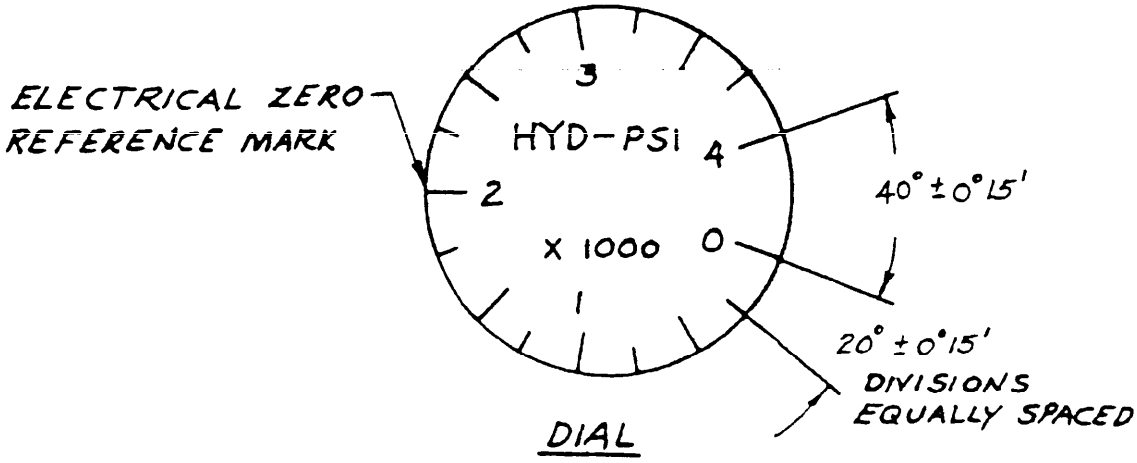
	Height of Length Inch ±1/64	Width of Line or Graduation Inch ±0.005	Color No.
Numerals 0, 1, 2, 3, 4	3/32	0.0213	37875
Graduations at 0, 1,000, 3,000, 4,000	7/64	0.030	37875
Graduation at 2,000	3/32	0.030	37875
250 psi graduations	1/16	0.015	37875
500 psi graduations	7/64	0.015	37875
Lettering "HYD-PSI"	5/64	0.015	37875
Lettering "X 1,000"	5/64	0.010	37875
Background of Dial	----	-----	37038

* 3.5.3.5 Electrical receptacle. The indicator shall be provided with an electrical receptacle as shown on Figure 2.

3.5.3.6 Indicator weight. The weight of the indicator shall not exceed 0.50 pound.

3.5.3.7 Wiring diagram. The indicator wiring to the electrical receptacle shall be in accordance with Figure 2.

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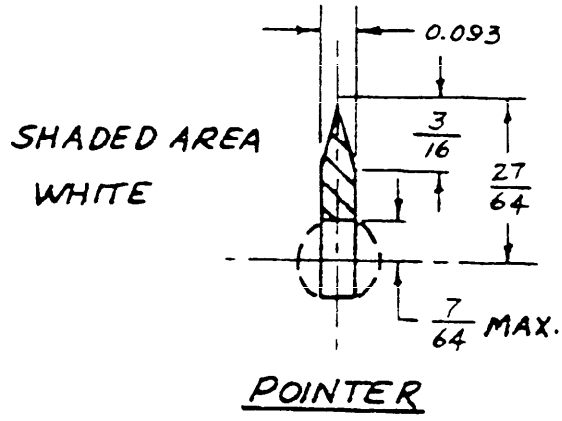


FIGURE 1. DIAL AND POINTER

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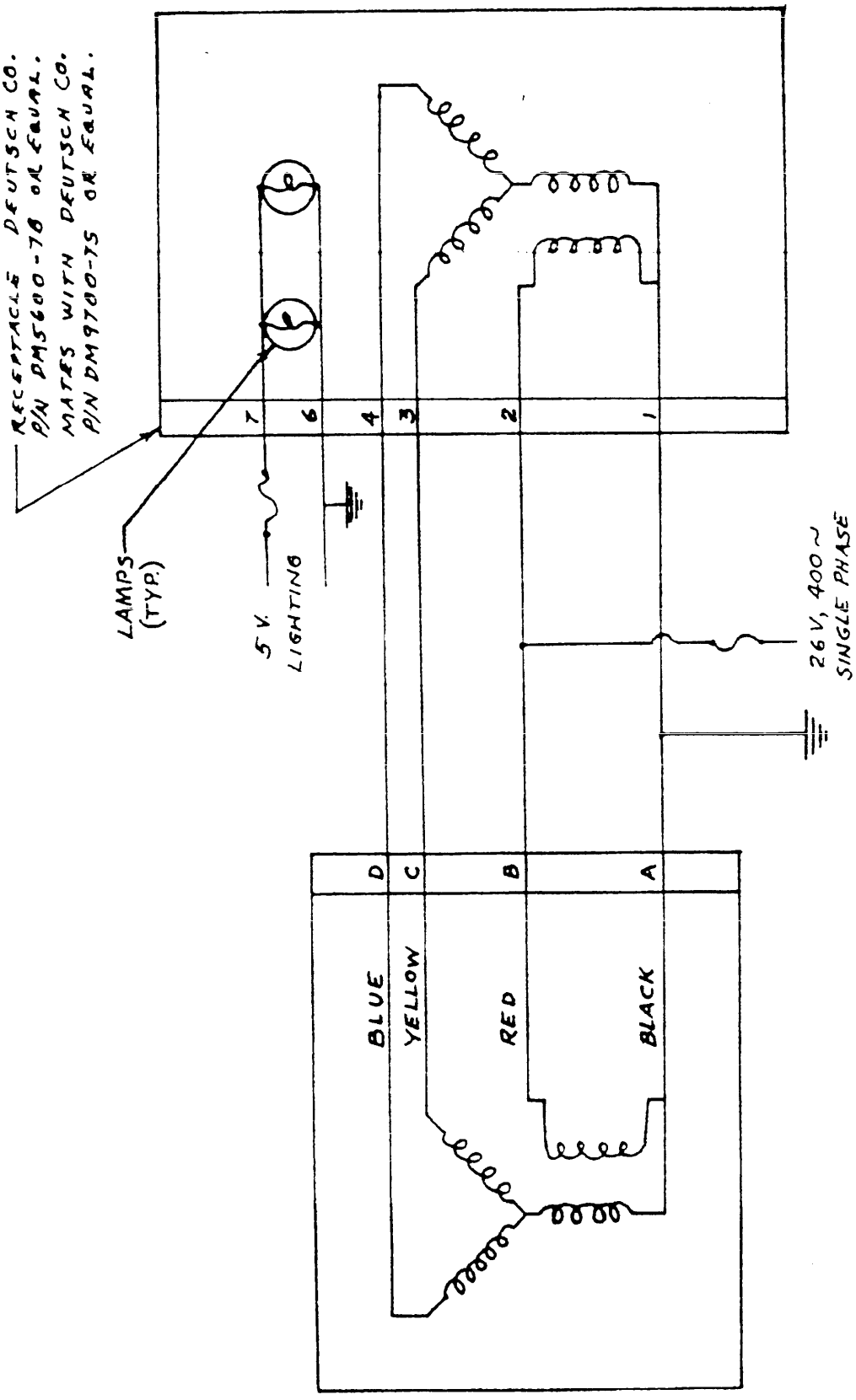


FIGURE 2. WIRING DIAGRAM

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3.5.3.8 Synchro shaft rotation. The rotation of the indicator synchro shaft, as viewed from the dial face, shall be clockwise for an increasing pressure indication.

* 3.5.3.9 Performance. The transmitter and indicator shall perform satisfactorily within the limits specified in Table I and Table II after being subjected to test methods of 4.4.

TABLE I

Pressure Transmitter

Test Pressure PSI	Tolerance ± PSI			
	Room Temp	Vibration Overpressure Seasoning	Low Temp High Temp (70°C) Alt, Endurance	High Temp (145°C)
0	50	75	75	100
500	50	75	100	125
1,000	50	75	100	125
1,500	50	75	100	125
2,000	50	75	125	150
2,500	50	75	125	150
3,000	50	75	125	150
3,500	75	100	125	150
4,000	75	100	125	150

TABLE II

Pressure Transmitter Errors

Test	Tolerance ± PSI
Friction	
Room Temperature	50
High Temperature (70°C)	50
High Temperature (145°C)	75
Low Temperature	75
Position Error at 2,000 psi	50
Leak	25
Vibration	
Pointer Oscillation	60

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* 3.5.3.10 Step input response. The time required for the indicator to properly respond to a step input equivalent to a 2,000 psi pressure change shall not exceed 2.0 seconds with not more than two overshoots.

* 3.5.3.11 Voltage and frequency variation. The indicator shall withstand voltage and frequency variations from 22 to 30 volts and 320 to 480 Hertz (Hz) without exceeding the scale error and friction tolerances specified in Tables III and IV.

TABLE III

Pressure Indicator Tolerances
All Scale Error Tests

PSI	Tolerance ± PSI	PSI	Tolerance ± PSI
0	50	3,000	50
500	50	3,500	50
1,000	50	4,000	50
2,000	50		
2,500	50		

TABLE IV

Pressure Indicator Errors

Test	Tolerance ± PSI
Position error at 2,500 psi	25
Friction	
Room Temperature	50
High Temperature	50
Low Temperature	75
Vibration	
Pointer Oscillation	50
Pointer Variation	50

3.5.4 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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submitted for qualification shall be identified . . . All test samples

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the pressure indicating system shall be as follows:

* 4.2.1 Preproduction. Preproduction testing of the transmitter and indicator shall be as specified herein and in test methods of MIL-T-5350.

* 4.2.2 Quality conformance. Quality conformance inspection shall be in accordance with MIL-T-5350.

4.3 Test condition.

4.3.1 Transmitter. Unless otherwise specified, the transmitter shall be tested with the longitudinal axis in a horizontal plane.

4.4 Test methods.

4.4.1 Individual tests.

4.4.1.1 Transmitter. The transmitter errors for the individual tests shall not exceed the applicable tolerances specified in Tables I and II.

4.4.1.2 Indicator. Indicator errors for the individual tests shall not exceed the applicable tolerances specified in Tables III and IV.

* 4.4.1.3 Step input response. The indicator pointer shall be positioned at approximately 0 psi pressure. With excitation power off, the test transmitter shall be positioned to the equivalent of 2,000 psi. Excitation power shall then be applied and the time required for the indicator to indicate 2,000 ±100 psi shall not exceed 2.0 seconds. The pointer shall move smoothly and shall not oscillate nor shall it overshoot more than two times.

* 4.4.1.4 Voltage and frequency variation. The indicator shall meet the test specified in 4.4.1.3 and the scale error and friction tolerances specified in Tables III and IV when the voltage and frequency are varied from 22 to 30 volts and 320 to 480 Hz or any combination thereof.

4.4.2 Sampling plan A.

4.4.2.1 High temperature (transmitter). The transmitter shall be subjected to a temperature of 145° ±2° Centigrade (C) for a period of 4 hours after which it shall be subjected to the scale error and friction tests. The scale error shall not exceed the tolerance for high temperature (145°C) specified in Table I, and the friction error shall not exceed the tolerance for high temperature (145°C) specified in Table II.

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4.4.3 Sampling plan B.

4.4.3.1 Vibration (transmitter). The transmitter shall be subjected to a vibration test in accordance with procedure XII of MIL-E-5272, except that curve II of Figure 3 shall be used in lieu of curve A of MIL-E-5272. A pressure sufficient to produce 80 ±5 percent deflection shall be applied to the transmitter during this test. Vibration errors shall not exceed the tolerances specified in Tables I and II, and no damage to the transmitter shall result from this test.

4.4.3.2 Vibration (indicator). The indicator shall be subjected to a vibration test in accordance with procedure XII of MIL-E-5272, except that curve I of Figure 3 herein shall be used in lieu of curve A of MIL-E-5272. This test shall be conducted with power applied and an input resistance sufficient to provide a pressure indication of 80 ±5 percent of the indicator range. The vibration errors shall not exceed the tolerances specified in Tables III and IV, and no damage to the indicator shall result from this test.

4.4.3.3 Extreme high temperature exposure (transmitter). The transmitter shall be subjected to a temperature of 145° ±2°C for a period of 4 hours. The temperature shall then be raised to 161° ±2°C for a period of 30 minutes. This cycle shall be repeated six times for a total of 24 hours at 145°C and 3 hours at 161°C. The transmitter shall then be tested for scale and friction errors at 145° ±2°C. The scale error and friction error shall not exceed the tolerances specified in Tables I and II, respectively. The transmitter shall be examined and there shall be no evidence of failure.

4.4.3.4 Leak test (transmitter). A pressure of 3,000 psi shall be applied to the pressure connection. The connection tube shall be completely sealed at a point with 2 inches of the connection. For a period of 1 minute, the pointer shall not change its position by more than the tolerance specified in Table II. This test may be combined with the transmitter scale error test.

4.4.3.5 Damping (transmitter). A pressure sufficient to produce full scale deflection of the pointer shall be applied to the transmitter. When the pressure is released suddenly, the time required for the pointer to traverse from 4,000 to 500 psi shall be noted and shall be within 1.0 to 3.0 seconds. This test shall be conducted with hydraulic fluid conforming to MIL-H-5606 at a temperature of -55° ±2°C.

4.4.3.6 Overpressure (transmitter). The transmitter shall be subjected to an overpressure of 1,000 psi (5,000 psi applied pressure) for a period of 10 minutes. Not less than 1 hour following this test, the transmitter shall be tested for errors on increasing pressures only. The scale error shall not exceed the tolerances specified in Table I.

4.4.3.7 Endurance (transmitter). The transmitter shall be subjected to 100,000 cycles of pressure applications to produce an indication of 2,500 to 3,500 psi and back to 2,500 psi at a rate of 30 ±5 cycles per minute (cpm). Not less than 1 hour after the pressure cycling, the transmitter shall be tested for scale errors at room temperature. There shall be no damage or excessive wear.

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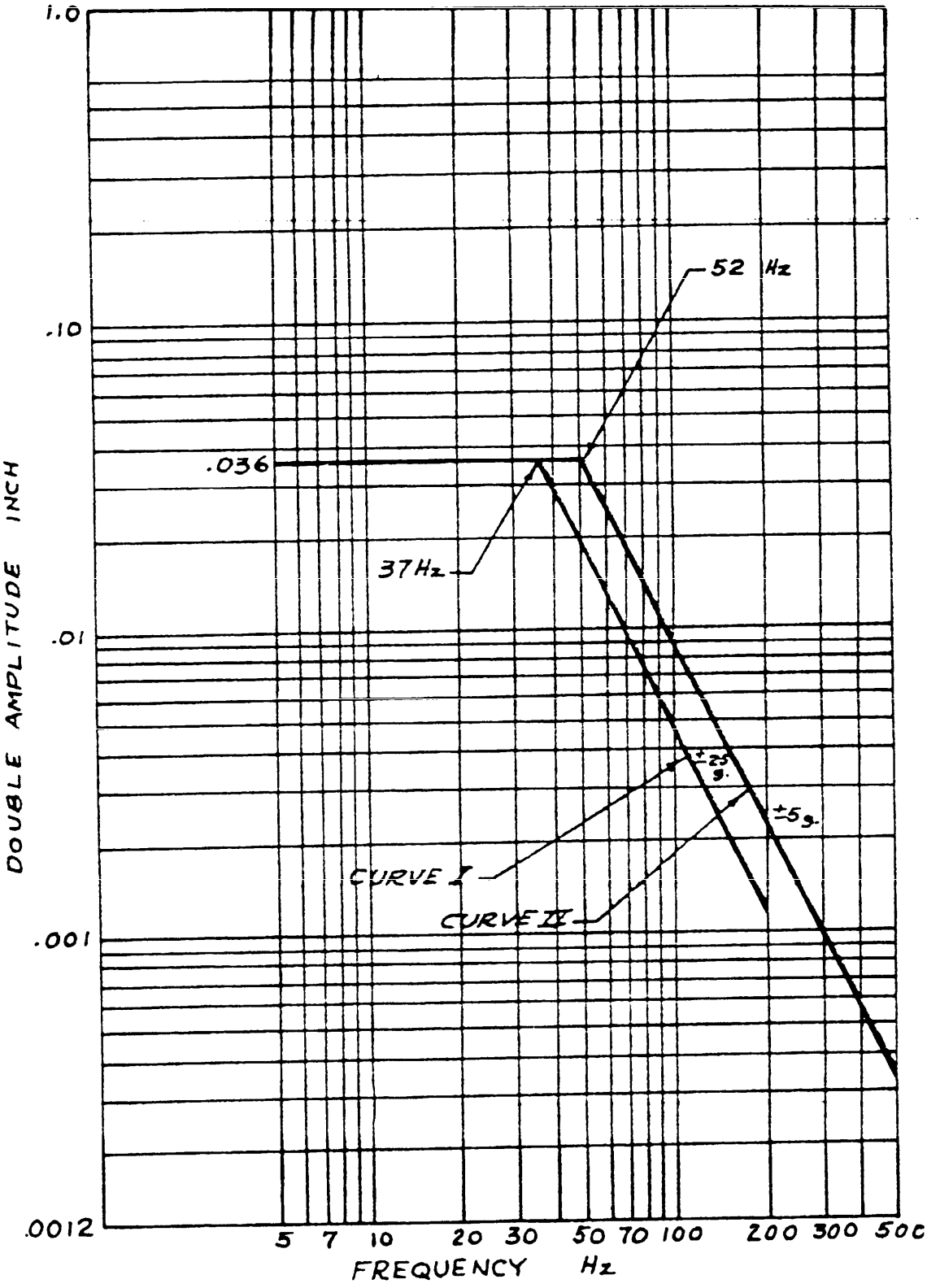


FIGURE 3. RANGE CURVES FOR VIBRATION TESTS

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4.4.3.8 Salt spray (transmitter). After exposure to the specified salt spray, the transmitter shall be examined for corrosion and then subjected to the sealing test. No scale or friction error test will be required.

4.4.3.9 Case pressure release (transmitter). The pressure blowout device shall be tested, either under simulated increasing case pressures or by actual injection of pressure into the transmitter case. The blowout device shall operate at a pressure of 80 ±25 psi.

* 4.4.3.10 The indicator shall be subjected to the tests specified in 4.4.1.3 and 4.4.1.4.

4.4.4 Reliability (system). The pressure indicating system shall be subjected to 1,500 hours of electrical operation. Pressure shall be applied to produce an indication of 2,500 to 3,500 and back to 2,500 psi at a rate of 30 ±5 cpm. During this time, there shall be a minimum of 6,000 cycles of full-scale indication (0 to 4,000 psi). This test shall begin when pressure is applied. After 4 hours of pressure cycling, the pressure shall be turned off and shall remain off for a 15-minute period. The pressure-on pressure-off procedure shall be repeated until the specified 1,500 hours of operation have been accomplished. In case of malfunction, the malfunction shall be recorded and the test interrupted while repairs are made. The test shall then be continued from the point of interruption. Upon the completion of each 250 hours of cycling operation, the indicator shall be tested for scale error. The scale error readings shall not differ by more than 1 percent of full scale from those obtained at room temperature previous to the test. No recalibration shall be required to pass the error test. This test may be interrupted for normal weekend shutdowns. The MTBF of the indicator, which shall be at least 750 hours, shall be determined from the following equation. The contractor shall submit a report to the procuring activity which shall contain test procedures, details of any malfunctions, and the corrective action taken.

$$MTBF = \frac{1500N}{M}$$

Where N = Number of units tests
M = Number of malfunctions

* 4.5 Inspection of the preservation, packaging, packing and marking for shipment and storage. Inspection and testing to insure compliance with Section 5 herein shall be accomplished in accordance with Section 4 of MIL-P-116.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

* 5.1.1 Level A. Each pressure indicating system shall be preserved and packaged in accordance with MIL-P-116, Method III and placed in a snug-fitting container conforming to Grade W6C, PPP-B-636.

* 5.1.2 Level C. Each pressure indicator system shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. The supplier may use his commercial practice providing it meets this requirement.

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* 5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

* 5.2.1 Level A. Pressure indicator systems preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to ~~PPP-B-636~~. The grade of each shipping container, determined by the gross weight and size, shall comply to the applicable special requirements table of the specification. Interior cushioning, blocking, bracing and waterproofing shall be in accordance with MIL-STD-1186. Closure and strapping shall be in accordance with the container specification. Containers shall be of uniform shape and size and of minimum cube and tare.

* 5.2.2 Level B. Same as Level A, except that waterproofing is not required and domestic containers may be used.

* 5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior type shipping containers in a manner that will insure safe transportation at lowest rates to the point of delivery. Containers shall comply with rules, or regulations of carriers as applicable to the mode of transportation.

* 5.3 Marking. In addition to any other marking required by order or contract, the interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The A/A27J-1 hydraulic pressure indicating system covered by this specification is intended for use in indicating aircraft hydraulic system pressure measurements.

6.1.1 The indicator can be used on instrument panels requiring case mounting in accordance with MS33639 by the application of an MS28042-4 mounting clamp.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether sampling plan B tests (special tests) of MIL-I-7057, sampling plan B tests of MIL-L-27160, and sampling plan B tests specified herein will be conducted.

c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

* 6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 82

Preparing Activity:
Air Force - 82

Project No. 6685-F431

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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